

Lab 02 Activity Sheet 01
Using MongoDB Compass/Atlas Database Setup

The objective of this tutorial is to set up the MongoDB database using Compass / Atlas to later be used for implementing CRUD operations (in Lab 02 Activity Sheet 02).

Lab Prerequisite:

Be familiar with the Week 2 lecture and resources prior to attending this activity.

There are three ways to access the MongoDB Database using Compass/Atlas.

To complete this lab activity you can choose any of the following ways to access the Database:

[Option 1: Store data in your PC using MongoDB Compass](#)

[Option 2: Store data on the cloud using MongoDB Atlas](#)

[Option 3: Store data on the cloud but use MongoDB Compass](#)

Following the above steps should result to atleast one database (`sample_restaurants`) and 2 collections (`neighborhoods` and `restaurants`).

After the Database is accessible (by completing any one of the above steps) you may proceed to Lab 02 Activity Sheet 02 where we will use the `restaurants` collection.

The `neighborhoods` will be used in later weeks.

Option 1: Store data in your PC using MongoDB Compass

Advantage: as the data will be stored on your PC, you will have more than 512MB space in the database

Disadvantage: can not access MongoDB Charts using Compass, will need to use Atlas to access Charts

1. Run the Compass App on your PC and connect to your PC's localhost server:

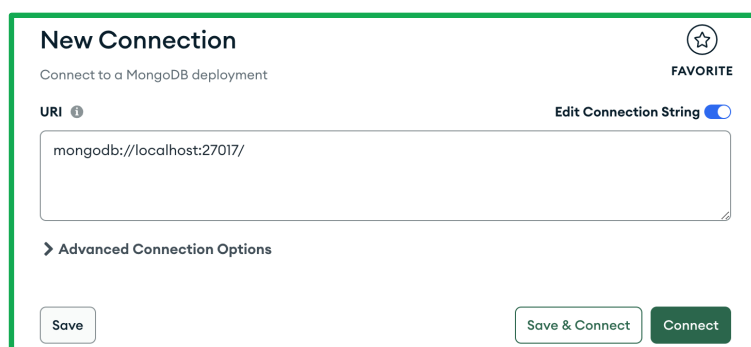


Figure: Connecting to the MongoDB localhost Server using MongoDB Compass

2. Create a database (`sample_restaurants`) and the first collection (`neighborhoods`):

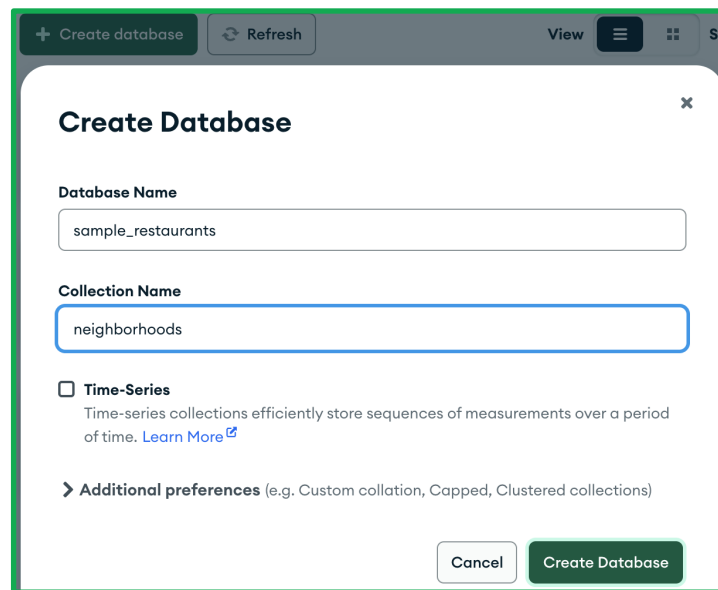


Figure: Creating a new database and collection using MongoDB Compass

3. Load the Sample Data for `neighborhoods.json` from Moodle > Week 3 > Lab Resources into the newly created `neighborhoods` collection:

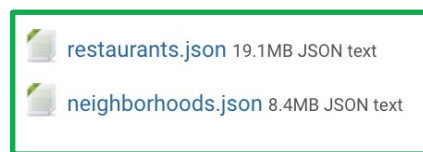


Figure: Moodle Files for loading Data



Figure: Two ways to load data from JSON files into the MongoDB Database using MongoDB Compass

4. Create the second collection called `restaurants` in the same `sample_restaurants` database and load the `restaurants.json` file from Moodle.



Figure: Creating a new collection using MongoDB Compass

Option 2: Store data on the cloud using MongoDB Atlas

Advantage: quick to startup after the cluster is created and doesn't require sample data to be loaded. Great alternative when Compass isn't working

Disadvantage: can only store upto 512MB (with the sample data taking up 79% (403MB) of the DB storage

1. Setup the Atas Account and Cluster using [MongoDB Installation Guide \(2023 Mac Windows Linux\)](#) > Cloud MongoDB Atlas Setup

2. Load the sample data on next section [MongoDB Installation Guide \(2023 Mac Windows Linux\)](#) > Checking MongoDB Atlas

3. Navigate to the sample_restaurants dataset to see the 2 collections containing the same data as the files in Moodle

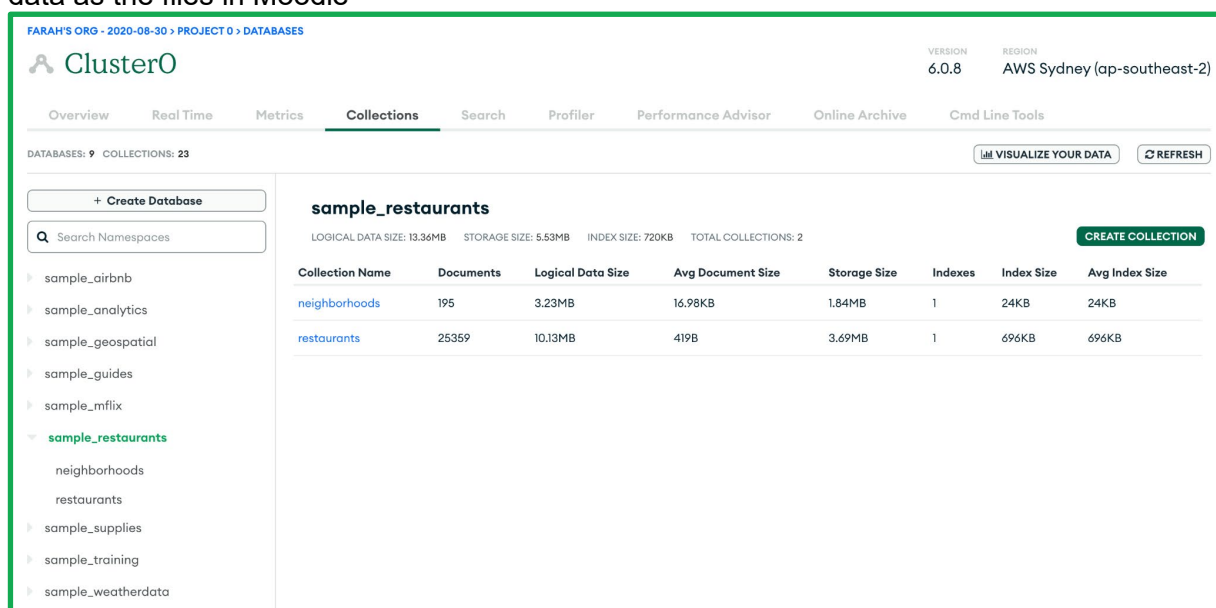


Figure: Sample Databases in MongoDB Atlas

Option 3: Store data on the cloud but use MongoDB Compass

Advantage: easy to switch to the local storage (Option 1) and doesn't require sample data to be loaded

Disadvantage: can only store upto 512MB (with the sample data taking up 79% (403MB) of the DB storage

Similar to [option 2](#):

1. Setup the Atas Account and Cluster using [MongoDB Installation Guide \(2023 Mac Windows Linux\)](#) > Cloud MongoDB Atlas Setup

2. Load the sample data on next section [MongoDB Installation Guide \(2023 Mac Windows Linux\)](#) > Checking MongoDB Atlas

3. In Atlas click on Connect:

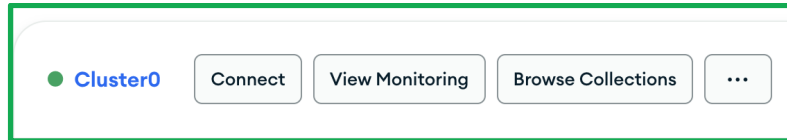


Figure: MongoDB Atlas Cluster Options

4. In the connection window select compass and copy the connection string:

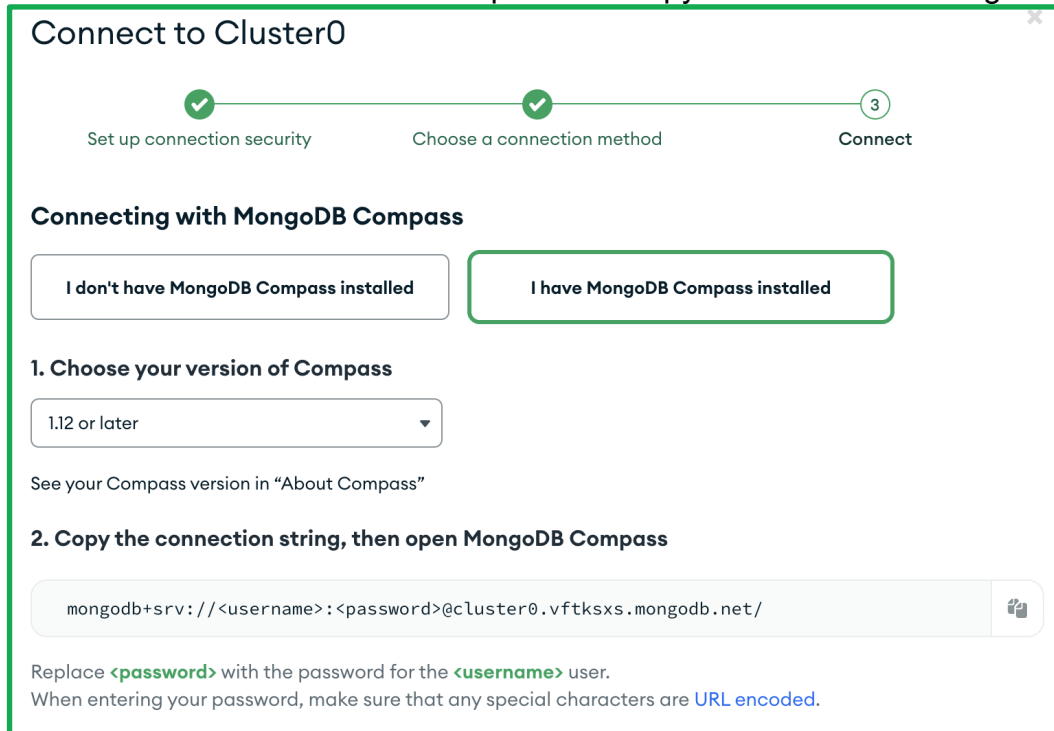


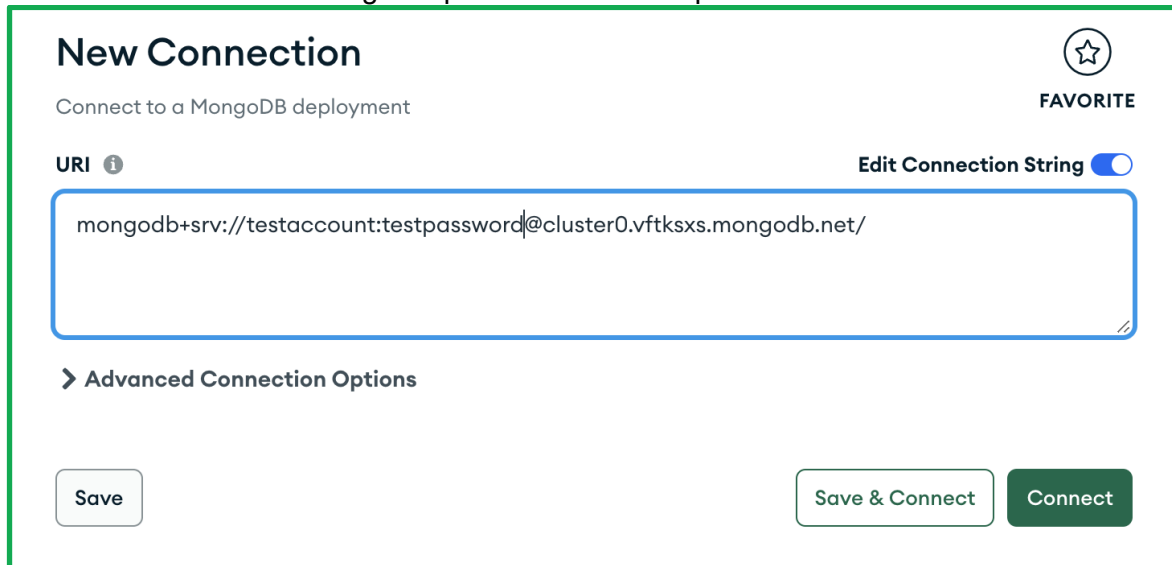
Figure: MongoDB Atlas Connect > Connecting to MongoDB Compass Option

The username and password can be set in
 Atlas > Security > Quickstart
 (in the password avoid using : / ? # [] @)



Figure: MongoDB Atlas > Security > Quickstart

5. Paste the connection string and password into Compass:



New Connection FAVORITE

Connect to a MongoDB deployment

URI ⓘ Edit Connection String ☒

mongodb+srv://testaccount:testpassword@cluster0.vftksxs.mongodb.net/

➤ Advanced Connection Options

Save Save & Connect Connect

Figure: Connecting to the MongoDB Atlas Server using MongoDB Compass

Note: You may need to toggle Edit Connection string

6. Once connected you should see the list of sample databases:



Figure: Sample Databases in MongoDB Compass connected to the MongoDB Atlas Server

THE END